the most critical indicator in the investigation on aeration savings, follow by the NH_3-N and $CBOD_5$ (appendix C, Figure C.3).

6.4 Results and discussion

6.4.1 Steady-state conditions. In general, D.O. and NH_3-N satisfy their respective permit requirements in the different test conditions. For example, in the standard operating conditions (Figure 6.5), the strictest NH_3-N limit (2.5 mg N/L) and D.O. requirement (4 mg/L) can be achieved in about 20% and 35% of the total length of the tank, respectively

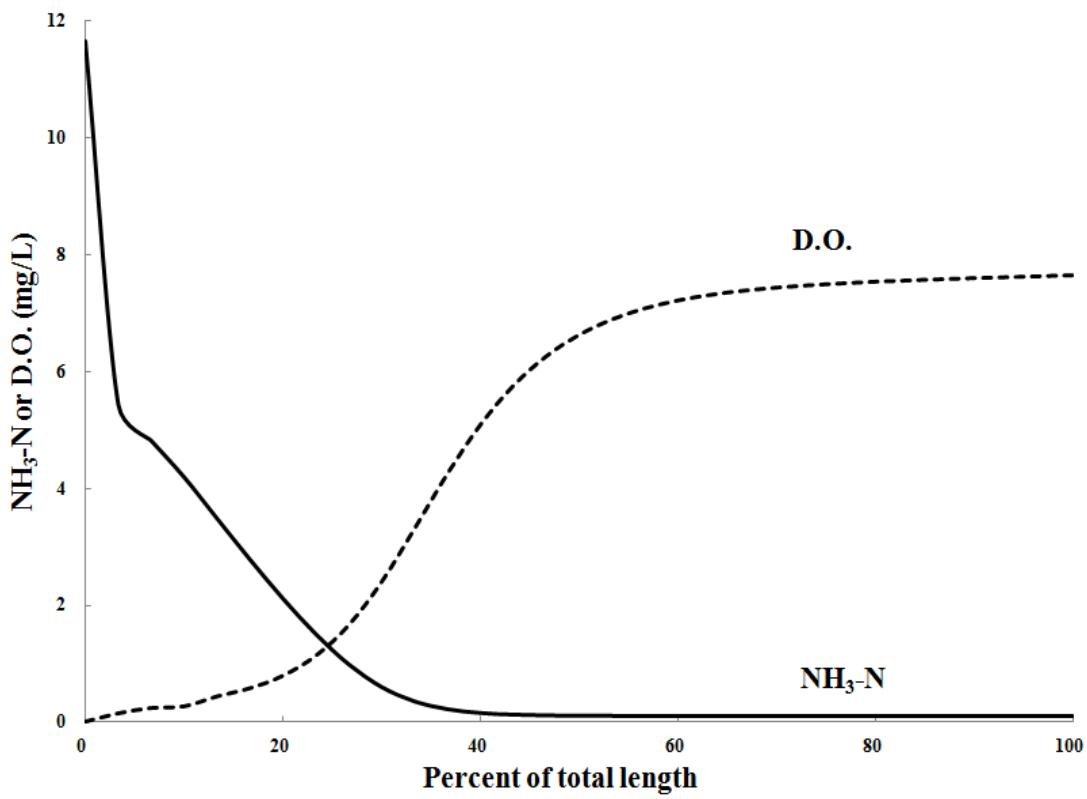

Figure 6.5. Simulated ammonia and D.O. concentration profiles along the aeration tank for standard operating conditions (total length of tank = 425 ft).

The other seven groups of steady-state conditions exhibit similar pattern in D.O. and NH_3-N profiles along the aeration. Low temperatures conditions (groups 2 and 6) result in slightly higher effluent D.O. concentrations. When TARP flow is 0 (groups 1 and 5), the D.O. concentration reaches a constant value more quickly.

Steady-state simulations were used to explore aeration requirements for each of the eight influent characteristic groups. For example, when the plant is operating under the defined standard conditions (group 3), about 50% aeration savings could be realized while still meeting the effluent requirement for D.O. (Figure 6.6).

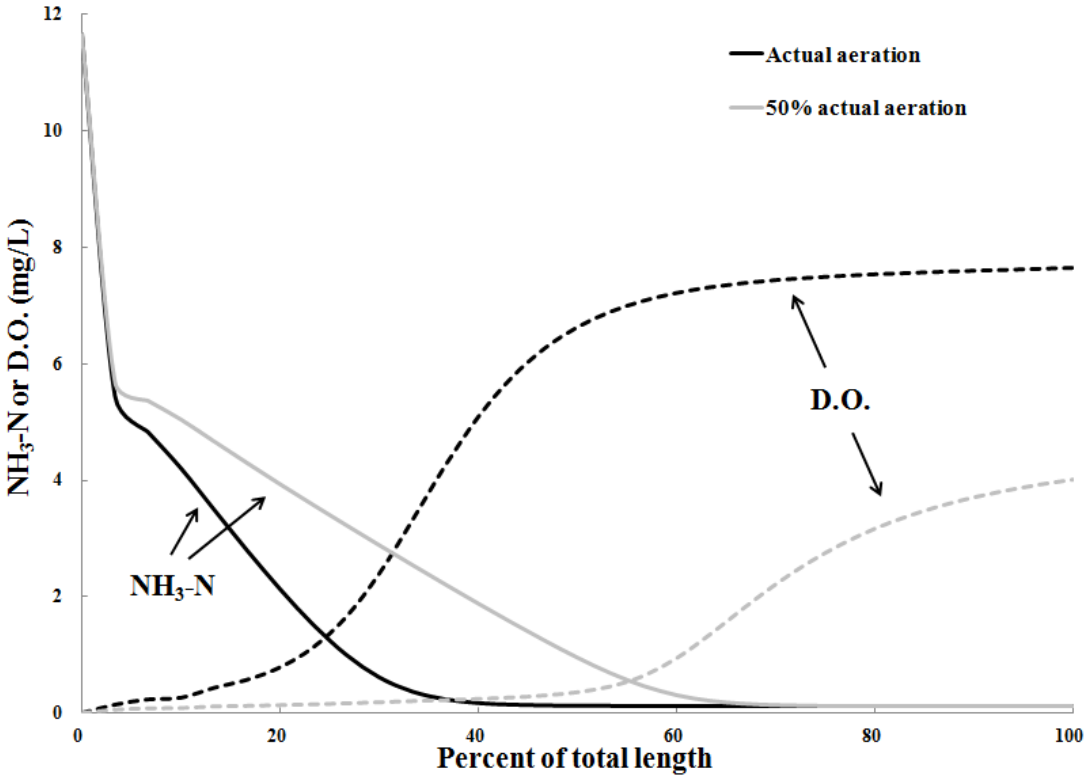

Figure 6.6. Change of D.O. and NH_3-N concentration profiles when 50% of the current average aeration (42.3 MCFD) is used during the standard conditions (total length of tank = 425 ft). The effluent D.O. just meets its permit requirement of 4 mg/L and NH_3-N in the effluent is still far under its limit.

It seems possible to meet the D.O. requirement with aeration savings ranging

from 47.4% to 66.5% among the eight groups; the smallest savings is in group 7 (precipitation, TARP flow, medium temperature), and the highest savings is in group 5 (precipitation, no TARP flow) (Figure 6.7). For operations constrained only by the ammonia limit, it is possible to achieve much higher aeration savings, ranging from 70.1% to 79.6%. Different amounts of aeration apply to each of the eight groups, and as a result although group 4 (no precipitation, TARP flow, high temperature) does not have the highest aeration savings (≈ 55.6%), it has the highest possible reduction of aeration (≈ 31.8 MCFD) because its actual aeration (≈ 57.3 MCFD) is the most. The lowest reduction of aeration (≈ 18.9 MCFD) may be obtained in group 6 (precipitation, TARP flow, low temperature), which has relatively low actual aeration (≈ 39.1 MCFD) and saving percentage (≈ 48.4%).

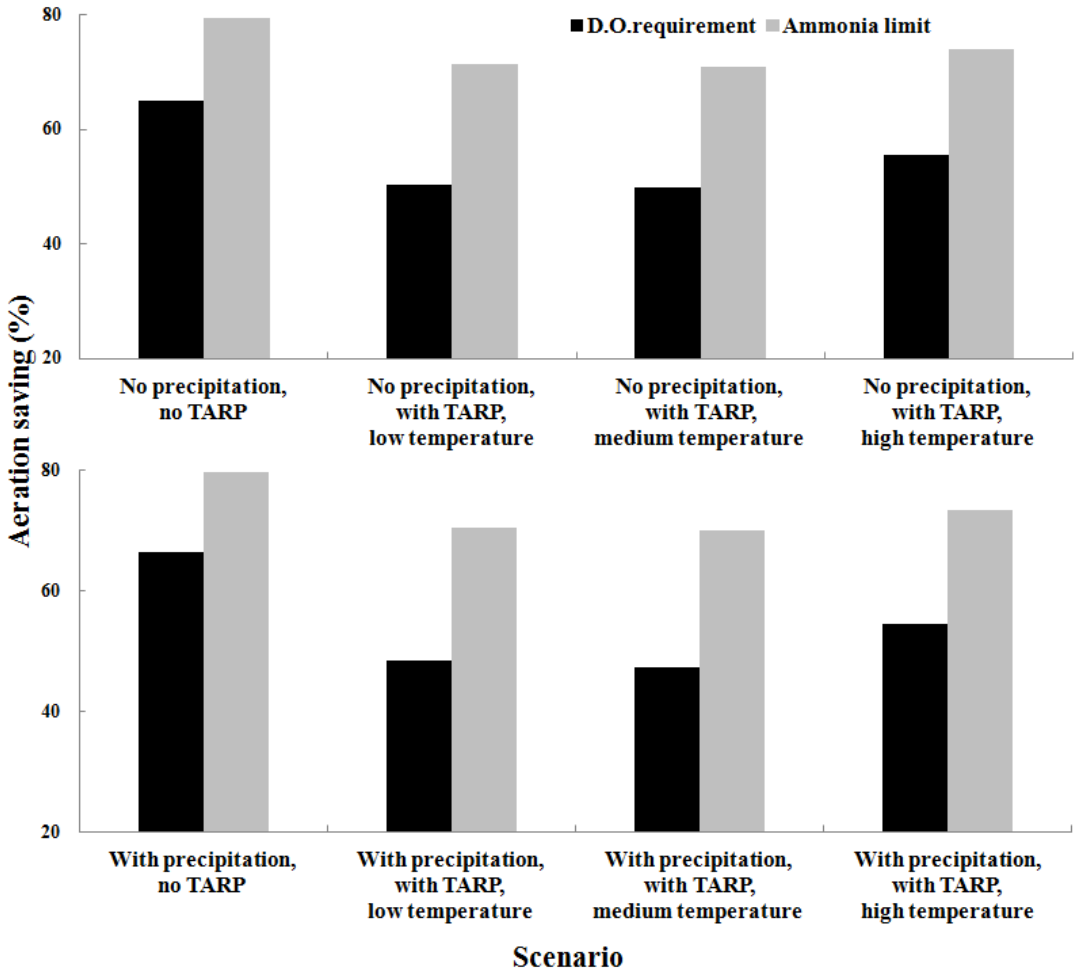

Figure 6.7. Possible aeration savings for eight groups of influent conditions. The process meets the permit D.O. requirement or ammonia limits in each case.

Overall aeration savings in steady-state conditions can be estimated based on the savings from each group and the frequency of each group above. The result suggests a 71% savings in aeration could be achieved, and the plant would still satisfy the strictest ammonia permit limit. Process constraints for D.O. are more stringent; however, a 50% decrease in aeration is still possible. Even when taking into consideration the limit on the suggested minimum mixing aeration demand, the plant could still decrease the current aeration amount by up to 50%, and meet all the permit requirements.

6.4.2 Transient scenarios. When an influent perturbation occurs, concentrations of ammonia, $CBOD_5$, and D.O. in the effluent can exhibit very different behaviors depending on the aeration rate.

As an example of WRP response to a storm event, the 100 MGD event (its peak flowrate is about 3.5 time of the average flowrate) (Figure 6.8(a)), shows that the D.O. profile exhibits little change throughout the perturbation. The profile projection on the lower face of the figure helps identify conditions where the D.O. falls below the permit requirement (4 mg/L); these conditions are limited to the first 35% of the length of the aeration tank. During the storm, rapidly increasing (t = 26 h) and decreasing (38 h) flows would require additional aeration or a slightly longer aeration tank to satisfy the D.O. permit. Compared to the D.O., the ammonia concentration profile has a bigger change during the perturbation because dilution substantially decreases the influent ammonia concentration (Figure 6.8(b)). The ammonia concentration can meet the permit limit within 20% length of tank all the time and within only 5% length of tank in peak flow during the perturbation. In response to the perturbation, the autotrophic biomass concentration decreases initially and then gradually recovers after the event (appendix C, Figure C.4). Compared to D.O. (99.5%) and ammonia (99.2%), which have almost completely recovered by the end of storm (48 h), autotrophic biomass has a slower recovery (91.1%).

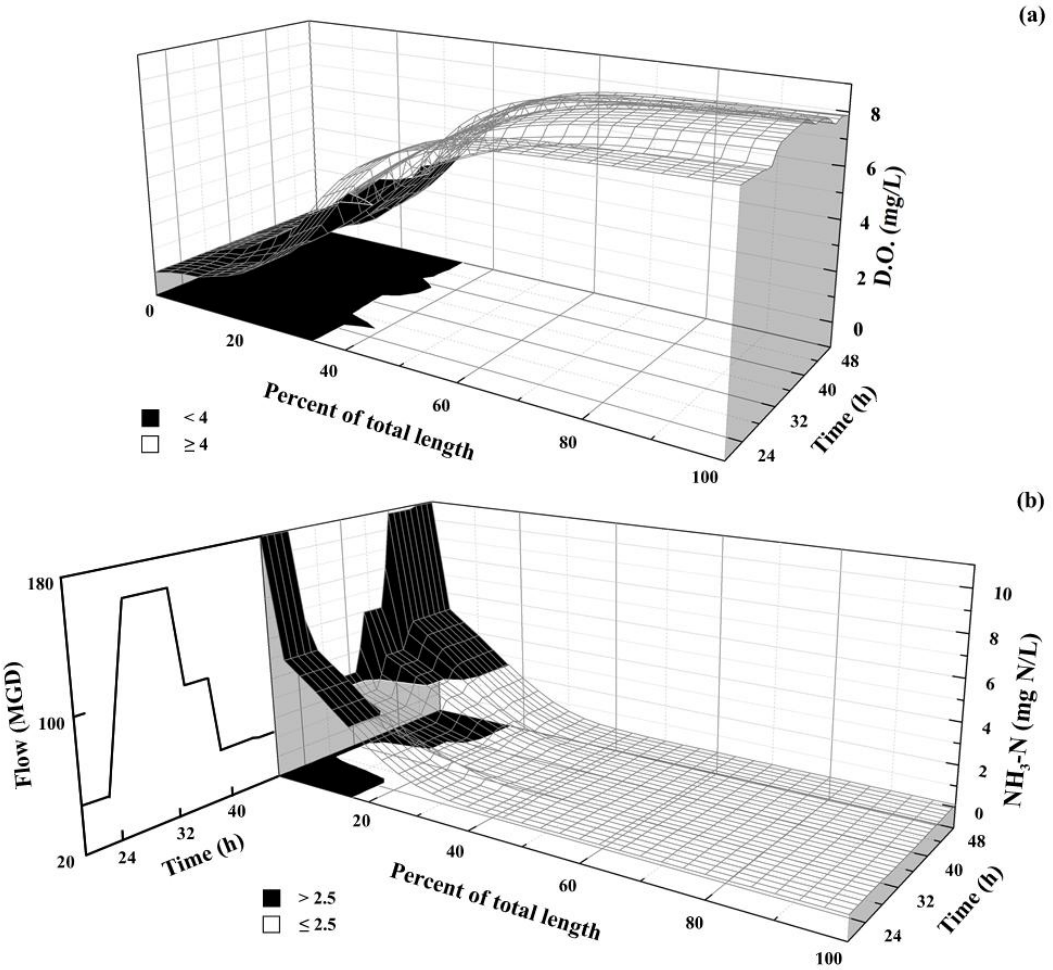

Figure 6.8. D.O. and ammonia concentration profiles in the aeration tank during a 100 MGD high probability storm event (total length of tank = 425 ft). The dark shading identifies conditions where the (a) D.O. and (b) ammonia do not meet effluent requirements. The 100 MGD flow pattern is shown in (b) ammonia for reference.

The system response to simulated storm events for the high and low probability perturbations at 60, 80, 100, and 120 MGD varied in amplitude but followed similar patterns. These differences can be more clearly seen in the effluent quality as described as the following paragraph.

As an example of comparing different flowrates, simulation results of the high probability perturbations show that a larger flow has a bigger impact on the effluent D.O.

and ammonia (Figure 6.9). The effluent D.O. concentration increases about 30 h into the perturbation and the system gradually recovers after about 50 h. The larger the magnitude of the perturbation, the greater the increase in D.O. and the longer it takes the process to recover. The effluent ammonia concentration exhibits almost a mirror-image response through the perturbation. Storm flow causes lower D.O. concentration and a small ammonia concentration peak at around 27 h probably because the increase in flow essentially short-circuits the treatment process, so the wastewater leaves the aeration tank before it has been treated sufficiently. Although 60, 80, 100, and 120 MGD storm events have different impacts on the effluent quality, the worst effluent quality (7.7 mg O_2/L and 0.11 mg NH_3-N/L) is still far from the permit requirements for D.O. ($\geq$ 4 mg/L) and ammonia ($\leq$ 2.5 mg N/L).

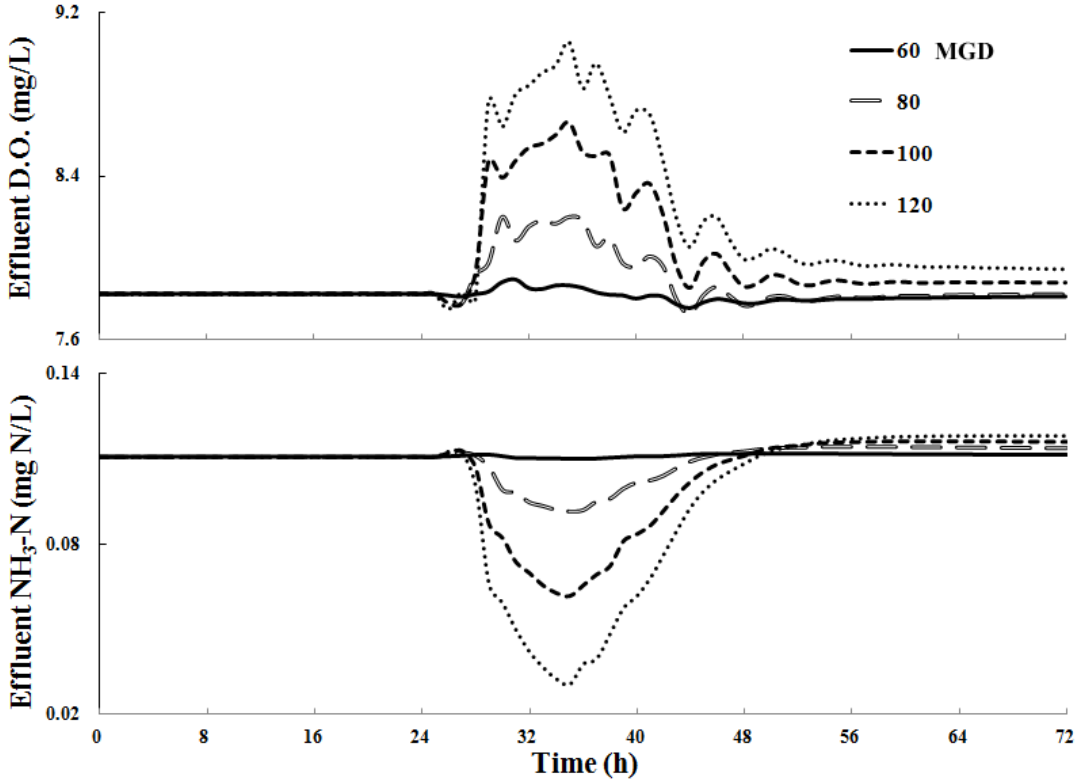

Figure 6.9. Response of effluent D.O. and NH_3-N concentrations to 60, 80, 100, and 120 MGD high probability storm events.

In summary, under the most common steady-state scenario, the Calumet WRP is able to satisfy all permit requirements within 35% of the total length of the aeration basin. When there is a mean perturbation (even for the rare – less than 2% of the time – 120 MGD flow), the Calumet WRP requires only 40% of the length of the aeration basin to manage the event. Even when the Calumet WRP faces the challenge of a low probability (< 1%) 100 MGD storm event, only half the length of the aeration basin is required to satisfy the treatment requirements. These results suggest that aeration at the Calumet WRP is often in excess, and that there is substantial opportunity to decrease the aeration supply.

The amount of aeration during a perturbation event has an effect on effluent quality; the D.O. profile exhibits greater fluctuations for lower aeration rates (Figure 6.10(a)). In contrast with the D.O., ammonia concentrations are much less sensitive to changes in aeration as long as the aeration is higher than 20 MCFD (Figure 6.10(b)). If aeration drops below 16 MCFD, however, the effluent ammonia concentration increases rapidly. When the D.O. concentration is too low (< 2 mg/L), it significantly inhibits nitrification (Gerardi 2002). Based on the D.O. requirement the minimum aeration requirements for the 60, 80, and 120 MGD high probability storm events are 20.6, 21.3, and 23.7 MCFD, respectively. Based on the ammonia limit these minimums fall to 11.7, 11.9, and 12.2 MCFD, respectively.

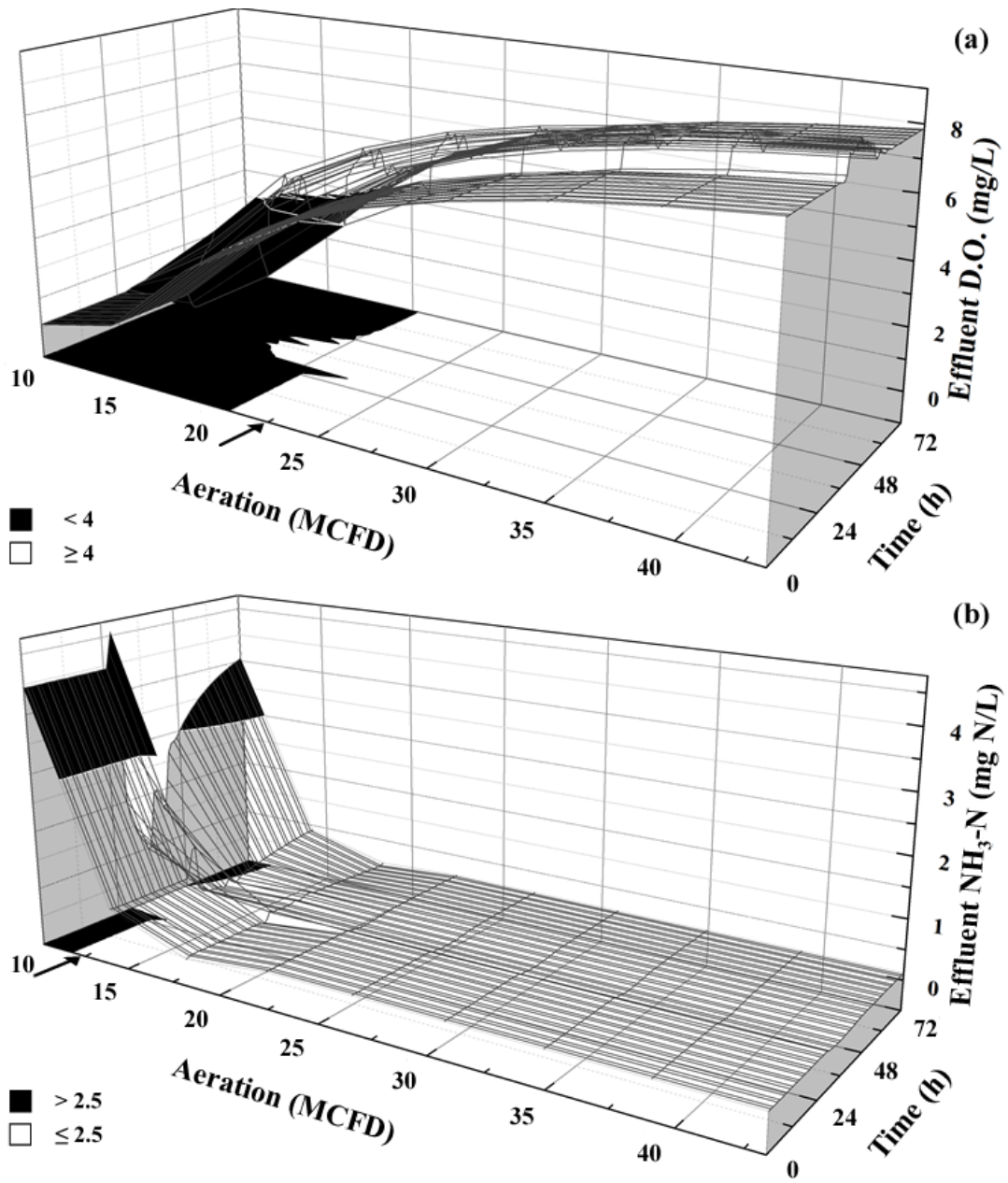

Figure 6.10. Change of effluent (a) D.O. and (b) ammonia concentrations as a function of the aeration rate during the 100 MGD high probability storm event. The darker shade identifies regions were the permit requirements would not be met.

Based on the ammonia limit, aeration could be decreased by over 70% for all high probability perturbations (Figure 6.11). The D.O. requirement is more limiting, with the

decrease in aeration ranging from just over 50% to just under 45%, as the influent flow is increased from 60 to 120 MGD. When especially challenging, low probability perturbations happen, the pattern of changes in effluent quality is similar but more aeration is needed to meet the permit requirements. For example, based on tests with the four simulated flows, relative to the high probability perturbations in each case the aeration savings was about 10% lower based on the ammonia limit. For the D.O. requirement, the saving is decreased to around 30%. Even though the plant requires more aeration to meet all permit requirements in the extreme perturbations, these results suggest that aeration could be decreased at least 29% and as much as 37% with no adverse effect on operations.

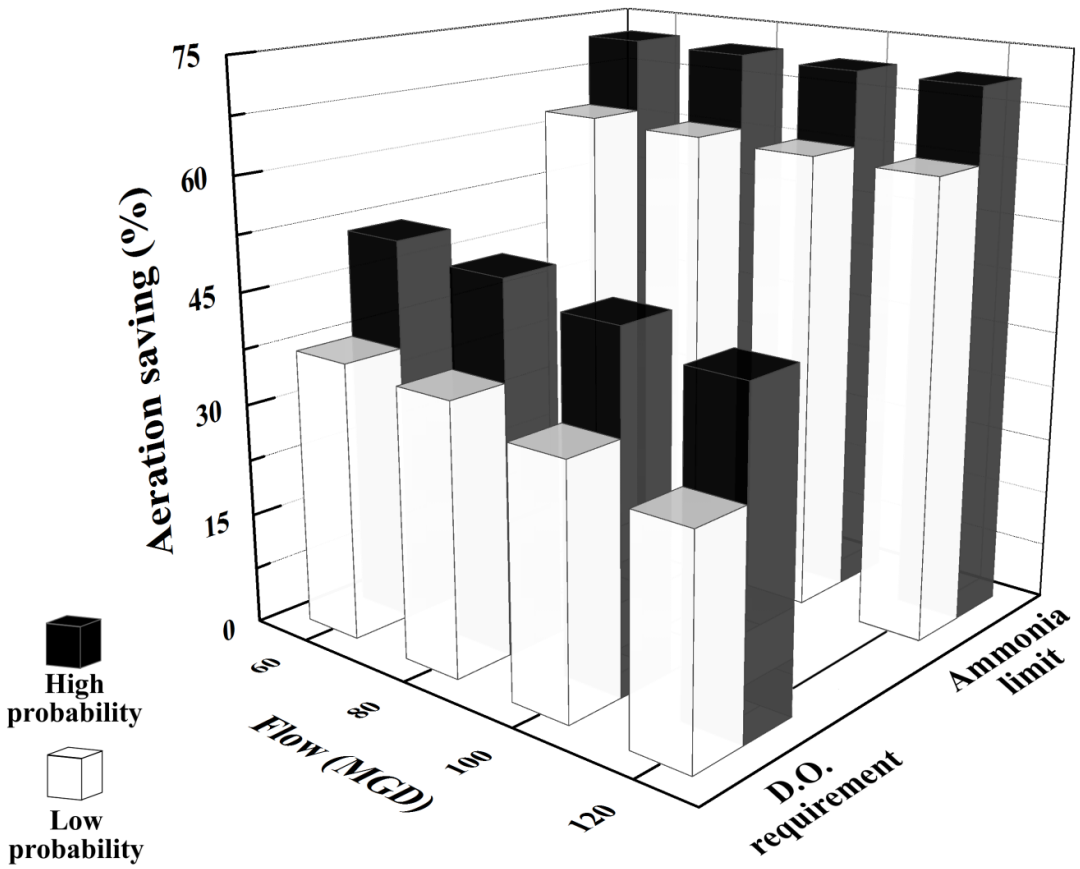

Figure 6.11. Theoretical aeration savings associated with the 60, 80, 100, and 120 MGD high probability and low probability storm events, respectively.

6.5 Overall evaluation. Historical data, steady-state profile simulations, and D.O. field measurements, all support the idea that the Calumet WRP frequently operates with excess aeration.

From the tests conducted under steady-state conditions, the least amount of energy savings occurs for an influent scenario that includes precipitation and TARP flow, and relatively low water temperature. Historical data indicate that the aeration rate was less under low temperature conditions, probably due to higher solubility of oxygen and slower kinetics at lower temperatures. During simulated storm events, the worst effluent

quality occurs at the highest flowrates when there is a sudden decrease in the hydraulic residence time (HRT) and biomass concentration. In other words, rather than an immediate rise, a step-by-step increase of flowrate from TARP may provide relatively higher effluent quality.

Based on mixing requirements (U.S. EPA 1989), the minimum level of aeration in tank E2 can be realized with about 48% of the current aeration rate. Generally, it is possible to reduce the amount of energy for aeration demand by 50% during steady-state conditions and the process can still satisfy all the permit requirements (Figure 6.12). Storm flows can have an adverse effect on the process, and larger storm flows can have a greater effect on the process and effluent quality. In average influent conditions, aeration can be decreased by about 49%. Even during a low probability perturbation (< 4%), energy savings of up to 35% appear feasible, which exceeds the amount (30%) proposed in the original hypothesis.

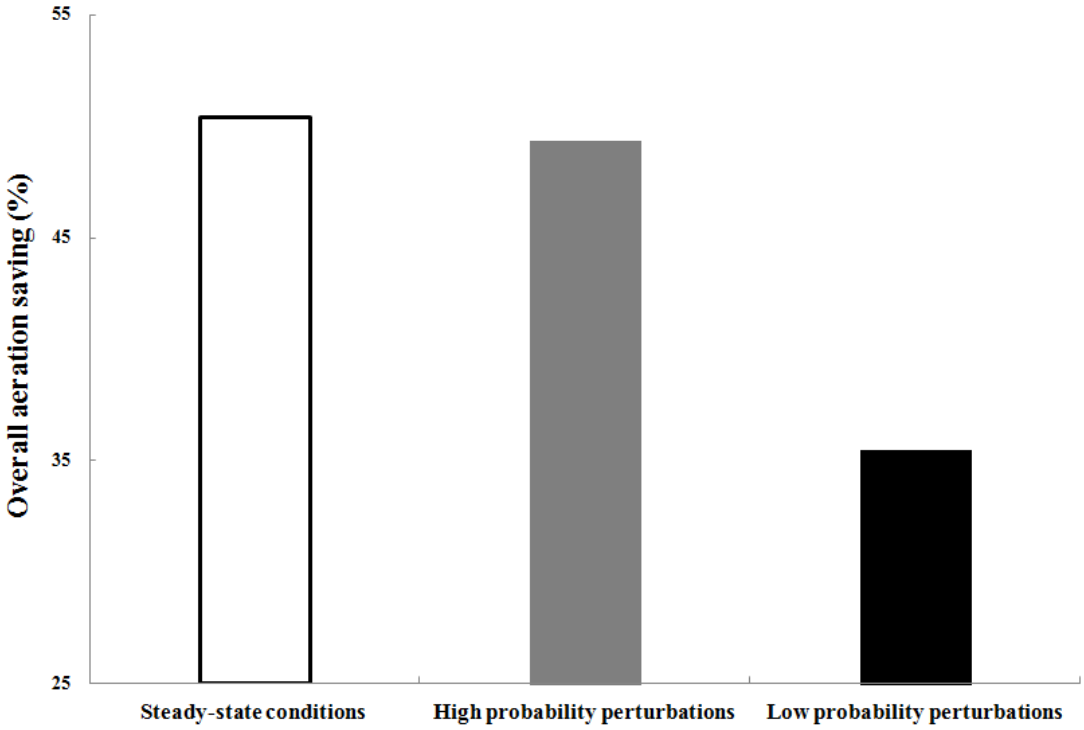

Figure 6.12. Comparison of overall aeration savings among steady-state conditions, high probability perturbations, and low probability perturbations.

Ten years of historical data suggest that the average daily electricity consumption for the blower is about 125.5 thousand kilowatt hour (tkwh) for the whole plant. Assuming a cost of 0.075 $/kwh in 2005 (MWRDGC 2005b), the annual cost for the blower is about $3.4 M. If these results for battery E2 apply throughout the Calumet WRP, the annual savings stemming from cutting the aeration by only 35% would be more than $1.2 M. At this reduced aeration rate the WRP should be able to meet effluent quality standards and maintain the resilience required to manage low probability perturbations.

The energy savings associated with decreased aeration is limited by the need to meet the D.O. permit requirement. Currently forced aeration takes place only in the aeration tank, so the D.O. in the outfall is about the same as the D.O. in the effluent from

the aeration tank. An alternative approach would be to provide just enough aeration to meet treatment requirements in the aeration basin, and use a post-aeration process to meet the effluent D.O. requirement. Because the oxygen transfer rate (KLa) is higher in clean water than it is in wastewater, less energy would be required to satisfy the D.O. requirement. To examine the α-factor (KLa ratio of wastewater to clean water), we considered the work by Henkel et al (2011), who reviewed more than 14 studies and reported that the α-factor can be correlated with MLVSS (MLSS) concentration and solid retention time (SRT). Applying this correlation to the Calumet WRP, we estimate that the α-factor is about 0.56 under typical operating conditions. In other words, there could be a 44% difference in the oxygen transfer efficiency between the aeration basin and the treated effluent. Addition of a post-aeration system could be an efficient way to further optimize the distribution of energy demand between meeting the ammonia limit and D.O. requirement.

The possibility of energy savings at the Calumet WRP suggests that other old WRPs, especially those with dated control techniques or lacking a post-aeration process, could also operate more cost-effectively. Although influent characteristics, operational conditions, and effluent permits can vary in different WRPs, implementation of more advanced control techniques, such as monitoring and prediction of influent conditions, should allow the old plants to reduce energy costs/CO_2 emissions in relatively steady-state influent conditions and still to keep their effluent quality even during huge perturbations.

6.6 Summary

In this chapter, the influence of both steady-state and two different kinds of perturbation conditions on the plant were examined via the WRP model, and the possible energy savings under steady-state and transient influent scenarios at the Calumet WRP were evaluated. Results can be briefly summarized as below:

- Effluent D.O. is the limiting factor that determines aeration requirements at the Calumet WRP. Considering steady-state conditions, mixing requirements, and influent perturbations, with a CPS installation up to 50% of the energy used for aeration could be saved and the process could satisfy all permit requirements.
- Higher influent concentrations of TKN and $CBOD_5$, and lower SS concentration can adversely affect effluent quality. Even when faced with these challenging conditions, however, relative to current conditions the average aeration rate can be reduced by more than 30% and the perturbations can still be managed effectively.

These results provide a clear primary picture of general energy saving at the Calumet WRP. The next step was to further obtain detailed results based more sophisticated influent scenarios. To do that, clustering and cross-tabulation methods were used to statistically classify influent conditions at the Calumet WRP. This approach is described in the next chapter, along with an evaluation of energy savings for different influent scenarios.